Chapter 4 — Stoichiometry

Board-style MCQs · PECTAA Chemistry 11 · Answer key at end

Practice MCQs

1. One mole contains:

(a) 6.02 × 10^{23} particles **(b)** 1 g particles **(c)** 22.4 particles **(d)** 6.02 × 10^{22} particles

2. Moles from mass are calculated by:

(a) n = m/M **(b)** n = PV/RT only **(c)** n = M/m **(d)** n = mV

3. Molar mass of water:

(a) 2 g mol^{-1} **(b)** 20 g mol^{-1} **(c)** 18 g mol^{-1} **(d)** 16 g mol^{-1}

4. Mass of 2 mol CO_2:

(a) 11 g **(b)** 88 g **(c)** 44 g **(d)** 22 g

5. Molecules in 0.5 mol O_2:

(a) 1.20 × 10^{24} **(b)** 1.00 × 10^{23} **(c)** 3.01 × 10^{23} **(d)** 6.02 × 10^{23}

6. Molar volume at STP is about:

(a) 24 dm^3 **(b)** 100 dm^3 **(c)** 1 dm^3 **(d)** 22.4 dm^3

7. Volume of 0.5 mol gas at STP:

(a) 44.8 dm^3 **(b)** 22.4 dm^3 **(c)** 11.2 dm^3 **(d)** 5.6 dm^3

8. Molar mass from STP density:

(a) 22.4 / density **(b)** density / 22.4 **(c)** density x 22.4 **(d)** density x 2

9. Molar concentration unit:

(a) g dm^{-3} **(b)** mol dm^{-3} **(c)** dm^3 mol^{-1} **(d)** mol g^{-1}

10. Moles in 250 cm^3 of 1 M solution:

(a) 1 **(b)** 2.5 **(c)** 0.25 **(d)** 250

11. Balanced equation gives:

(a) Mole ratio **(b)** Density **(c)** Colour **(d)** Atomic mass

12. Limiting reactant is:

(a) Left in excess **(b)** Catalyst **(c)** Consumed first **(d)** Product

13. Theoretical yield is:

(a) Experimental mass **(b)** Always zero **(c)** Impure mass **(d)** Calculated maximum

14. Percentage yield formula:

(a) actual/theoretical × 100 **(b)** actual - theoretical **(c)** actual x theoretical **(d)** theoretical/actual × 100

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15. Actual yield is usually less due to:

(a) More catalyst **(b)** Higher purity **(c)** Lower molar mass **(d)** Side reactions/losses

16. Law of conservation of mass requires:

(a) Unbalanced equation **(b)** No atoms **(c)** No moles **(d)** Balanced equation

17. Mass of 1 mol NaCl:

(a) 58.5 g **(b)** 100 g **(c)** 23 g **(d)** 35.5 g

18. Number of moles in 4 g He:

(a) 4 **(b)** 1 **(c)** 0.25 **(d)** 2

19. Number of atoms in 1 mol He:

(a) 6.02×10^{23} **(b)** 1 **(c)** 2 **(d)** 3.01×10^{23}

20. Stoichiometry is important in medicine for:

(a) Density only **(b)** Colour only **(c)** Smell only **(d)** Correct dosage

21. To find product mass, first convert given mass to:

(a) Volume only **(b)** Temperature **(c)** Density **(d)** Moles

22. Mole ratio comes from:

(a) Atomic numbers **(b)** States **(c)** Coefficients **(d)** Subscripts

23. Excess reactant:

(a) Is catalyst **(b)** Is product **(c)** Remains after reaction **(d)** Is fully consumed

24. Molar mass is numerically equal to:

(a) Density **(b)** Valency **(c)** Volume **(d)** Atomic/molecular mass in g

25. Avogadro number is named after:

(a) Amedeo Avogadro **(b)** Boyle **(c)** Dalton **(d)** Charles

26. 22.4 dm^3 of CO_2 at STP has mass:

(a) 16 g **(b)** 22 g **(c)** 44 g **(d)** 28 g

27. 0.1 mol NaOH mass:

(a) 10 g **(b)** 40 g **(c)** 0.4 g **(d)** 4 g

28. Concentration = moles /:

(a) Pressure **(b)** Temperature **(c)** Volume in dm^3 **(d)** Mass in g

29. Percentage yield cannot normally exceed:

(a) 50% **(b)** 10% **(c)** 200% **(d)** 100%

30. Stoichiometric calculations obey:

(a) Variable atoms **(b)** Definite proportions **(c)** No conservation **(d)** Random ratios

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31. Avogadro's number is:

(a) 22.4 mol^{-1} **(b)** 6.02 × 10^{23} mol^{-1} **(c)** 6.02 × 10^{22} mol^{-1} **(d)** 1.66 × 10^{-24} mol^{-1}

32. Molar mass is mass of:

(a) One mole **(b)** One atom **(c)** One gram **(d)** One litre

33. Number of particles equals:

(a) n/NA **(b)** V/NA **(c)** n x NA **(d)** m x NA

34. At STP, one mole gas occupies about:

(a) 1 dm^3 **(b)** 24 dm^3 **(c)** 22.4 dm^3 **(d)** 100 dm^3

35. Molar mass from gas density at STP:

(a) density x 22.4 **(b)** density x 273 **(c)** 22.4/density **(d)** density/22.4

36. Molarity is:

(a) mol dm^{-3} **(b)** mol g^{-1} **(c)** g dm^{-3} **(d)** dm^3 mol^{-1}

37. Dilution changes:

(a) Molar mass **(b)** Concentration **(c)** Avogadro number **(d)** Moles of solute

38. Balanced equation coefficients give:

(a) Atomic number **(b)** Mass ratio directly **(c)** Density **(d)** Mole ratio

39. Mass-mass calculation first converts mass to:

(a) Temperature **(b)** Moles **(c)** Volume **(d)** Density

40. Gas volume at STP from moles:

(a) V = n x 22.4 **(b)** V = 22.4/n **(c)** V = nRT **(d)** V = n/22.4

41. Theoretical yield is:

(a) Impure yield **(b)** Always zero **(c)** Actual laboratory yield **(d)** Maximum calculated yield

42. Actual yield is:

(a) Molar mass **(b)** Maximum calculated yield **(c)** Always 100% **(d)** Laboratory obtained yield

43. Percentage yield equals:

(a) actual/theoretical × 100 **(b)** theoretical/actual × 100 **(c)** actual x theoretical **(d)** actual - theoretical

44. Limiting reactant is:

(a) Catalyst **(b)** Consumed first **(c)** Solvent **(d)** Left over

45. Excess reactant is:

(a) Left after reaction **(b)** Always gas **(c)** Always solid **(d)** Consumed first

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46. Stoichiometry in medicine ensures:

(a) Correct dosage **(b)** Tablet shape only **(c)** Colour of tablet **(d)** Brand name

47. 0.5 mol HCl mass is about:

(a) 0.5 g **(b)** 18.25 g **(c)** 36.5 g **(d)** 73 g

48. 20 g NaOH is:

(a) 0.5 mol **(b)** 2 mol **(c)** 1 mol **(d)** 40 mol

49. 1 mol CO_2 contains:

(a) NA molecules **(b)** NA atoms **(c)** 2NA molecules **(d)** 22.4 atoms

50. Molar concentration uses volume in:

(a) mol **(b)** dm^3 **(c)** g **(d)** cm^3 only

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Answer Key with Brief Reasons

1. Answer: (a) 6.02 × 10^{23} particles — Avogadro number.

2. Answer: (a) n = m/M — Mass divided by molar mass.

3. Answer: (c) 18 g mol^{-1} — H_2O = 2 + 16.

4. Answer: (b) 88 g — M CO_2 = 44; 2 x 44.

5. Answer: (c) 3.01 × 10^{23} — Half Avogadro number.

6. Answer: (d) 22.4 dm^3 — One mole ideal gas at STP.

7. Answer: (c) 11.2 dm^3 — Half molar volume.

8. Answer: (c) density x 22.4 — M = d x molar volume.

9. Answer: (b) mol dm^{-3} — Moles per volume of solution.

10. Answer: (c) 0.25 — n = C x V in dm^3.

11. Answer: (a) Mole ratio — Coefficients give reacting moles.

12. Answer: (c) Consumed first — It limits product amount.

13. Answer: (d) Calculated maximum — From balanced equation.

14. Answer: (a) actual/theoretical × 100 — Efficiency of reaction.

15. Answer: (d) Side reactions/losses — Practical losses reduce yield.

16. Answer: (d) Balanced equation — Atoms are conserved.

17. Answer: (a) 58.5 g — Na 23 + Cl 35.5.

18. Answer: (b) 1 — M He = 4.

19. Answer: (a) 6.02 × 10^{23} — He is monatomic.

20. Answer: (d) Correct dosage — Exact amounts ensure safety.

21. Answer: (d) Moles — Use mole ratio next.

22. Answer: (c) Coefficients — Balanced equation coefficients.

23. Answer: (c) Remains after reaction — Not limiting.

24. Answer: (d) Atomic/molecular mass in g — Grams per mole.

25. Answer: (a) Amedeo Avogadro — Historical naming.

26. Answer: (c) 44 g — One mole CO_2.

27. Answer: (d) 4 g — M NaOH = 40.

28. Answer: (c) Volume in dm^3 — C = n/V.

29. Answer: (d) 100% — Actual cannot exceed theoretical.

30. Answer: (b) Definite proportions — Fixed composition and ratios.

31. Answer: (b) 6.02 × 10^{23} mol^{-1} — Particles per mole.

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32. Answer: (a) One mole — g mol^{-1}.

33. Answer: (c) n x NA — Mole-particle link.

34. Answer: (c) 22.4 dm^3 — Book molar volume.

35. Answer: (a) density x 22.4 — Mass of one mole.

36. Answer: (a) mol dm^{-3} — Moles per volume.

37. Answer: (b) Concentration — Adding solvent lowers concentration.

38. Answer: (d) Mole ratio — Stoichiometric ratio.

39. Answer: (b) Moles — Use n=m/M.

40. Answer: (a) V = n x 22.4 — Molar volume relation.

41. Answer: (d) Maximum calculated yield — From balanced equation.

42. Answer: (d) Laboratory obtained yield — Measured product.

43. Answer: (a) actual/theoretical × 100 — Efficiency of reaction.

44. Answer: (b) Consumed first — Controls product amount.

45. Answer: (a) Left after reaction — Not fully used.

46. Answer: (a) Correct dosage — Safe active ingredient amount.

47. Answer: (b) 18.25 g — 0.5 x 36.5.

48. Answer: (a) 0.5 mol — 20/40.

49. Answer: (a) NA molecules — One mole molecules.

50. Answer: (b) dm^3 — c=n/V.

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